

4/11/2026 Magnet & Spring Order

- Current Magnet Numbers:
 - Cylindrical: 11
 - Flat: 6
- Needed Mangets:
 - Cylindrical: 24 total, but might need more → **20 more?** (in case it breaks and valve sticks)
 - Flat:
 - 30 * 2 cm for the flat gelbelt side
 - $60 / 1.9 = 31.579$
 - 7 * 4 + 30 * 2 cm for around
 - $100 / 1.9 = 52.632$
 - too much – calculate more conservatively
 - assume we have 5 of the around magnets, 1 of the flat side magnet
 - we need $(29.389 * 4)$ for sides + $(27.4 * 2 + 300 * 2)$ for top = 772.356 of around magnets
 - $772.356 / 19 = 40.65$ magnets (since we have 5, **35 more**)
 - and 31.379 of flat gelbelt side magnets (since we have 1, **30 more**)
 - Result: We decided to order only for one side now
 - **15 cylindrical, 18 around, 15 flat**
- Spring:
 - Tension spring coefficient: 9.19 N / cm
 - $9.19 \text{ N / cm} * 0.224809 / (1/2.54) = 5.2476266 \text{ lbf / in}$
 - Tension spring 2.225 cm (0.875 in) long, with 1.5 offset = around 4 cm (1.5748 in) compression spring

Compression Springs



Squeeze these springs and they'll push right back to their original length. When compressed, they apply steady pressure to keep components securely in place. Release the pressure and they snap back instantly, just like the tip of a ballpoint pen.

Spring Steel—Strong and resilient, but best suited for dry environments since moisture will cause these springs to rust.

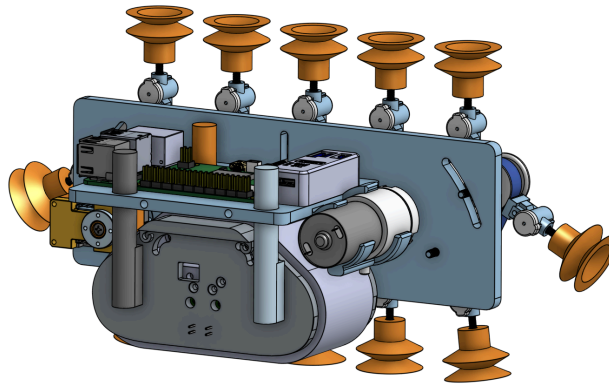
Corrosion-Resistant 302 Stainless Steel—Won't corrode from exposure to moisture and mild chemicals, including ammonia often found in cleaning solutions. These springs are easier to compress than steel, but can't handle as high a load.

Lg.	OD	ID	Wire Dia.	Compressed Lg. @ Max. Load	Max. Load, lb.	Spring Rate, lbf/in	End Type	Pkg. Qty.	Pkg.
Spring Steel									
0.375"	0.148"	0.12"	0.014"	0.185"	1	5.2	Closed	12	9657K611 \$7.50
0.5"	0.12"	0.09"	0.014"	0.263"	1.2	5.2	Closed	6	9657K237 10.39
1.25"	0.375"	0.305"	0.035"	0.595"	3.4	5.2	Closed	12	9657K155 9.45
2"	0.3"	0.236"	0.032"	0.856"	6	5.3	Closed and Ground	6	9657K385 8.04
3.5"	0.36"	0.276"	0.042"	1.83"	8.6	5.2	Closed and Ground	5	9434K116 5.53
4.5"	0.563"	0.437"	0.063"	2.75"	9	5.2	Closed	6	9657K221 12.28
Corrosion-Resistant 302 Stainless Steel									
0.75"	0.125"	0.093"	0.016"	0.54"	1.1	5.3	Closed	6	1986K46 4.29
1.375"	0.313"	0.251"	0.031"	0.785"	3	5.2	Closed	6	1986K63 4.38
1.375"	0.36"	0.288"	0.031"	0.785"	3.1	5.3	Closed	6	1986K755 10.97
1.5"	0.45"	0.404"	0.038"	0.715"	4.1	5.2	Closed and Ground	5	9435K111 6.75
1.75"	0.219"	0.163"	0.028"	1.47"	1.4	5.3	Closed	6	1986K62 4.47
2.5"	0.48"	0.39"	0.045"	1.19"	6.7	5.2	Closed	6	1986K68 11.97

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- Among these, I chose closed and ground since it is better to stay straight (to give less friction to the beams), and stronger in terms of maximum load (max load in the smaller inner diameter springs are too small)

04/13/2026 First Design Printing

- Current version:



- Problems of the first printed parts

- ☒ ~~Beams getting jammed~~

- metal shaft?
 - **More offset between the beams and the top plate for now**
 - For later: compression springs
 - Result: the compression springs have inner diameter of 10.262 mm, so 0.2 mm offset would be 10 mm. and we can have 0.4 mm offset with the top plastic if the 0.4 mm offset version works good. They are not going to experience a lot of shear in the springs, so 0.2 mm offset should be okay? the beams can be smaller, if that is helpful, but the spring diameter is not going to decrease, so no need to decrease if not necessary.
 - is bigger diameter beam better?
 - more surface area → more jamming
 - bigger diameter → more support, less chance of failing
 - how thick then?
 - I guess for now 1 cm is fine, for later, we can decrease, or I can mention about decreasing the diameter later
 - Actual Result: Make the beam smaller so that it can minimize the distance between the driving plate and the gelbelt sideplate. The constraint is the spring width

- ☐ The pump mount

- The pump mount is pretty sturdy, with good offset. we can take it out just by extending the PLAs a little bit
 - Problem:
 - doesn't look good
 - one side of it fell off during printing
 - needs A LOT of supports to print
 - print them in side ways?

- ☒ ~~Printability~~

- Make the extrusions for the spring holders separatable for both gelbelt side plates and the driving plates

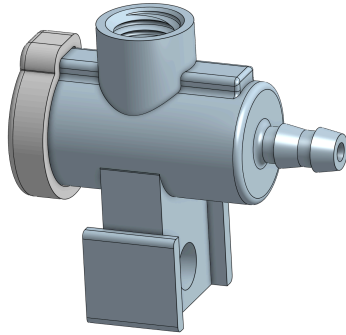
- Put the spring mounting hooks at the end of the extruded so that it is easier to print
- ☐ U2D2 should be mounted in some other ways, because it didn't have the screw hole
 - The holes that I supposed as the screw holes were not actually screw holes
 - Double sided tapes?
- ☒ ~~Relocate the U2D2 since its USB is in the way of the beams~~
 - Moved the top plate so that the beams are not in the way of the usb
- ☒ ~~Decrease the gap between the gelbelt sideplate and the driving plate~~
- ☒ ~~Make a hole for USB and other cables in the driving plate~~
- ☐ Think of a way to divide up the pump vacuum into two separate driving plates
- ☒ ~~The rotary joint is not secure~~
- ☒ ~~More offset for the top plate (I didn't consider I attached the top plate to the driving plate instead of the side plates, and therefore the offset should be made between the top plate and the gelbelt)~~
- ☒ ~~Make the beam screw to the gelbelt side plate from the other side (beams side) so that we can put heat insert to the gelbelt sideplate, not the beams~~
 - because if we want to attach from the gelbelt sideplate, we need to put a hole through the cylindrical support of the gelbelt sideplate, which is possible, but I think screwing from the beam is easier)
- ☒ ~~Bigger screw holes, smaller heat insert holes~~
- ☒ ~~The spring assemblies are hard since the space is tight and need to make sure the springs do not come off when screwing the top plate. better method?~~
 - Spring hooks in opposite directions
- Plan:
 - ☒ ~~Iterate the design~~
 - ☐ Implement the suspension system using compression springs
 - compression springs do not need the beams sticking out at the first place because the beams can go in the gelbelt sideplate. But I am not sure if that's the most efficient design (it might be the most pretty design though)
 - ☒ ~~modify the pneumatic valves~~
 - chain connection
 - bigger diameter?
 - tube connection?
 - ☒ ~~Print the version that I can show them. The current version is too incomplete~~
- Pneumatic valve thing to change
 - ☒ ~~Chain connection~~
 - ☒ ~~tube connection~~
 - ☒ ~~Cylinder inner diameter bigger~~
 - I increased the diameter by 0.2 (so 0.2 offset from each side of the cylindrical magnet)

☒ ~~suction cup connection~~

- should it be threaded?
- how did we end of having the junction? did I thread?
- I guess the thread for the junction was good. so I just copy pasted the same thing
- I increased the connection part extrusion +1 mm so that it threads more securely + less air leak

• Result:

- Pneumatic Valve:



04/15/2026 Second Printing (iterated with tension spring, not compression)

- ☒ ~~Note: if printing the top plate sideways because of the motor mount, the beam tolerance should be 0.4 mm since the holes are going to be squashed~~

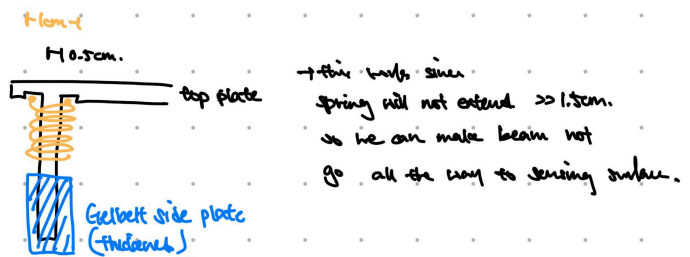


- ☐ The screw holes can be even bigger. now it's very tight
- ☒ ~~the springs are wrong. I used the stronger spring (7749N108) for design, but we should use 9.19 N spring (which is thicker)~~
- ☒ ~~beams don't get stuck → good~~
- ☒ ~~The spring holders should definitely be modified. it comes off so easily~~
- ☒ ~~only thing now is the offset~~

• Questions

- why do car suspensions use compression not tension springs?

- Safety – compression springs breaks less catastrophic than tension springs, allowing drivers to pull over safely
- Space constraint with damper
- handles extreme impacts better
- lightweight, cheaper
- what is better with compression springs in our design
 - more compact, simple
 - does it make it shorter (cm closer to wall)?
 - I don't think so. the gelbelt and the suction cups have the same relative distance to the wall anyway and the top plate is also the same relative distance to the gelbelt
 - easier to make the offset of 1.5 cm
 - because tension spring, the drive plate should attach to the bottom of the spring, which makes the drive plate inevitably extend very long, but compression spring can have the top part attached to the driving plate, which is a lot easier
- For compression spring design, we could also do this



- ☐ Dent in the top plate to make sure spring doesn't move around.
- ☐ The gelbelt side plate can be thickened
- ☐ this way the whole thing is narrower
- ☐ I don't know how it will work for 1 cm spring OD though

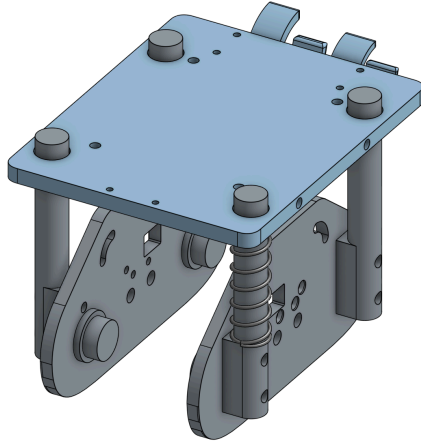
- For this design, when the beam is 1 cm, the gelbelt sideplate should be very thick. The tradeoff is that the gelbelt sideplate where it has holes for the beams are going to be weaker since the thickness is going to be small, and it's hard to print. Also, the spring attachment point on the gelbelt sideplate is going to be at the top of the gelbelt (unless we make another dent for the spring, which also risks gelbelt sideplate getting thicker & locally thinner near the holes), which increases the offset of the center of mass. so I don't think it is worth designing like this.

4/16/2026 Meeting Notes

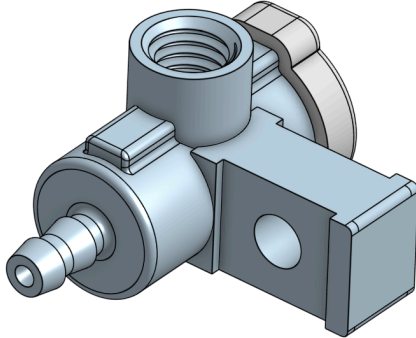
- ☒ ~~Modify the design for compression spring~~
- ☐ When the springs arrive, print and test
- ☐ Print and test the new pneumatic valve

- ☐ the acrylic method for silicone o-ring?
- ☒ 90-degrees
- ☒ make it shorter
- ☐ Cut / Mill the vented socket head screws

- Design for Compression Spring



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- Didn't do anything special
- Pneumatic Valve



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- 90 degrees chain connection, threaded for socket head screw, increased inner diameter, barbed tube connection
- Consider making the silicone o-ring process simpler
- attach the silicone orings to the magnets rather than the pneumatic valves.
- make silicone o-rings smaller than the magnet diameter
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